

Sensing Blocks

Learning Objectives

- Explore Scratch sensing blocks for input detection
- Use distance, color, and key-press sensors in programs
- Create interactive projects that respond to user actions

Engage (5-7 minutes)

A farmer uses senses to understand the world: seeing if clouds gather, hearing if rain falls, feeling if the soil is moist. Scratch sensing blocks give sprites similar abilities — to detect touches, colors, distances, and key presses, like digital senses for virtual characters.

Explore (10-12 minutes)

Students test sensing blocks: touching mouse-pointer, touching color, distance to sprite, and key pressed. Each block returns true or false, like a farmer's yes/no decisions based on observations. They see how sensing blocks connect to conditional logic.

Explain (10-12 minutes)

Sensing blocks detect the program's environment. "Touching?" checks if a sprite contacts something. "Distance to" measures how far apart objects are. "Key pressed?" detects keyboard input. "Color touching?" checks color contact. These blocks are the computer's sensory organs, providing information about the world around them.

Elaborate (8-10 minutes)

Students create a game where a sprite follows the mouse pointer, changing color when it touches the edge. Like a goat following its shepherd, the sprite uses sensing to stay connected to user input, creating responsive interaction.

Evaluate (5-8 minutes)

Can students use at least three different sensing blocks? Can they create a project where sprites respond to keyboard input? Do they understand how sensing blocks provide program input?